Gravitational field strength

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a gravitational field to predict the force on a test mass.

SUCCESS CRITERIA - I can...

1. Represent field direction with arrows.
2. Use g=F/m and F=mg consistently.
3. Distinguish source field from the mass placed in it.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “field” and “gravity” to explain your answer. Sentence starters are optional.

field: a description of an influence at each location

gravity: an attractive interaction between masses

radius: distance from the center to the edge of a circle or sphere

01 Warm-up: bring back an earlier idea

A **3 kg** object is in a field of **5 N/kg**. **Give its mass** and weight, and distinguish the two quantities.

Word help: weight = the gravitational force on an object

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $g = F_g/m = GM/r^2$.
Outside a spherical source of mass M; r is distance from its center. Field direction is toward the source.

Worked reasoning: A **2 kg** test object feels **6 N** toward a source. The local field is g=**6/2**=**3 N/kg**. A **4 kg** test object at that point feels **12 N**.

Your turn: 1. **Draw the field arrow** and the two possible force arrows.
2. Which quantity belongs to the location rather than the test mass?

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

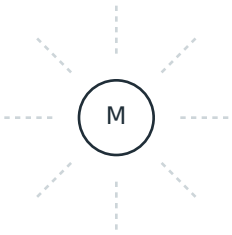
Test mass (kg)	Force (N)	g (N/kg)
2	6	___
4	12	___

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

1. **Draw field arrows** at three locations around the source.
2. Which way would a small test mass accelerate?

Sentence starter: In my work, I used ... to show ... . I labeled it ... .



Add arrowheads to show field direction.

The center marks the source; add arrowheads on the surrounding lines to show field direction.

Marzano check - circle: ① ② ③

04 Apply the physics

At one location, the gravitational field is **4 N/kg**. **Complete the force table** for three test masses. Does changing the test mass change the field at that location?

Sentence starter: I notice ... . This supports ... because ... .

Mass (kg)	Force (N)
1	___
2	___
3	___

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

Gravitational field strength

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A student says a heavier test mass falls faster because it feels more gravitational force. Use F=mg and a=F/m to respond.

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A **4 kg** cart moves left at **6 m/s**. Forces are **38 N** right and **6 N** left. **Find acceleration** and whether its speed initially rises or falls.
Draw the forces on the chosen object and **label their sources**.

Sentence starter: I found ... [unit]. This value means ... in this situation.

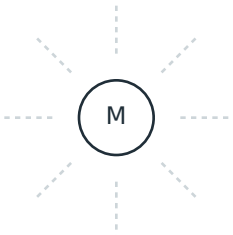
Add and label force arrows.

Marzano check - circle: 1 2 3

07 Exit check

A **2 kg** test mass feels **10 N**. **Find g** and the force on **4 kg** there.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Add arrowheads to show field direction.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Two test masses experience forces in a **1**: **3** ratio at the same point. Create possible masses and force values for a **5 N/kg** field.

Sentence starter: I notice ... . This supports ... because ... .

Location	Test mass (kg)	Field (N/kg)	Force (N)
1	____	____	____
2	____	____	____

Marzano check - circle: 1 2 3